

- 9 B** Graph G_1 for path from S to Q means that the two stars have different masses (since there is no point where resultant force is zero). Graph G_3 shows that star X has a smaller mass than star Y (the values of g at start is lesser than at the end). The point of $g = 0$ is closer to X. Since the spacecraft is accelerating, it is possible for the time duration from P to point of $g = 0$ to be approximately equal to the time duration from point of $g = 0$ to R.